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Explainable AI (SOW-BKI266)

Explainable AI: Comparing LIME and DiCE for Gender Bias Detection in Income Prediction

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Introduction

Automated systems are increasingly deployed for important decisions, such as hiring and credit scoring, yet these models risk amplifying existing biases which are often reflected in their training data (Barocas et al., 2023, p.7). The Adult Income Dataset serves as a vital diagnostic tool for shedding light on these systemic inequalities, as models trained on it frequently yield lower predicted incomes for women and minorities.

Explainable AI (XAI) acts as a vital socio-technical bridge to provide transparency, ensuring algorithmic decisions rely on legitimate reasoning rather than black-box output. Two broad paradigms have developed as dominant approaches to local explanations; that is, explaining individual predictions rather than global model behaviour. The first is feature attribution, most prominently realised by LIME (Local Interpretable Model-Agnostic Explanations), which explains individual predictions of any classifier by fitting a locally interpretable surrogate model around each instance of interest (Ribeiro et al., 2016). The second is counterfactual explanation, which instead asks: what minimal change to the input would have produced a different outcome? DiCE (Diverse Counterfactual Explanations) apply this approach by generating a diverse set of such hypothetical instances, optimising jointly for proximity to the original input and diversity across the counterfactual set (Mothilal et al., 2020).

While LIME is highly regarded for its model-agnostic approach, recent theoretical analyses reveal critical vulnerabilities, particularly regarding the instability of its feature attribution score across multiple runs. As demonstrated in Garreau and von Luxburg (2020), LIME's theoretical guarantees are fragile, with poor hyperparameter choices leading to inconsistent results for the exact same data point. Slack et al. (2020) further demonstrate the perturbation-based methods such as LIME can be systematically persuaded: a biased classifier can be scaffolded so that its real- world predictions remain discriminatory, while its post-hoc explanations appear entirely harmless. This finding exposes a fundamental limitation of feature attribution as bias-detection tool; it diagnoses what the model says it uses, not necessarily what it does use.

Counterfactual models offer a partial remedy: because they query the model directly with modified instances, they are harder to scaffold against in the same way. Yet standard counterfactual generation, as in Wachter et al. (2017), does not optimise for fairness it may still produce recourse suggestions that rely implicitly on protected attributes or their proxies. Goethals et al. (2023) address this gap with PreCoF, a metric that uses counterfactual explanations to detect not only explicit bias where a model directly uses a sensitive attribute but also implicit bias, where correlated proxy attributes disadvantage a protected group.

Ultimately, this literature leaves a persistent gap: while both LIME and DiCE have been independently validated for their interpretability and resourse generation, they are rarely compared directly regarding their ability to uncover implicit gender bias in income prediction models. This report addresses this gap by investigating whether feature attribution or counterfactual generation more reliably diagnoses a model's underlying dependence on protected attributes and their proxies.

Methods

LIME generates perturbed instances around a data point of interest, varying its input features to observe how the model's predicted output changes. By fitting a simple interpretable model to these perturbed samples, LIME identifies which features most strongly influence the prediction for that specific instance (Ribeiro et al., 2016). However, because this perturbation process relies on random sampling, LIME suffers from inherent instability, meaning different runs on the exact same data point can yield inconsistent feature importance scores (Garreau & von Luxburg, 2020). Furthermore, this reliance on localised approximations makes the method vulnerable to adversarial scaffolding, allowing fundamentally biased models to mask their discriminatory behaviour by presenting harmless explanations on perturbed inputs (Slack et al., 2020).
On the other hand, DiCE shifts the interpretability framework from scoring feature importance to generating “what-if” scenarios, effectively asking what minimal input change would flip the model’s prediction (Wachter et al., 2017). It achieves this by taking an original data point and optimising a loss function for proximity to the original instance, ensuring the hypothetical profile stays as close to the original as possible. To ensure users are given multiple viable options rather than a single strict path, DiCE enforces diversity among its suggestions using Determinantal Point Processes, a mathematical technique that ensures that counterfactual scenarios are distinct from one another (Mothilal et al., 2020). However, DiCE does not natively constrain counterfactuals to be fairness-aware, meaning suggested changes may still implicitly rely on protected attributes or their proxies (Goethals et al., 2023).

Finally, LIME and DiCE serve complementary roles in an algorithmic audit. LIME is best utilised as a diagnostic tool to expose exactly which feature the model is weighing heavily on, though it hides how easily those decisions can be changed. In contrast, DiCE reveals the actual boundaries of the model’s logic by showing what must change to alter the outcome. By applying both methods, an auditor can evaluate not just what the model claims is important, but what actually derives its final predictions (Slack et al., 2020).

Results

A Random Forest classifier was trained on the Adult Income Dataset, achieving an accuracy of 86% on the held-out test set, confirming the model performs well enough to be a reliable basis for explainability analysis.

```python
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

# separate feature from target variable
X = df.drop('income', axis=1)
y = df['income']

# split data into train and test sets(80/20 split)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train a Random Forest classifier on the Adult Income Dataset
# This is our black-box model that LIME and DiCE will later explain
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)

#  accuracy check
accuracy = accuracy_score(y_test, rf_model.predict(X_test))
print(f"Model accuracy: {accuracy:.2f}")
```

✓ [4] 8s 261ms

```
Model accuracy: 0.86
```

Figure 1 Random Forest classifier training and accuracy score on the Adult Income Dataset.

LIME explanations were generated for individual instances to identify which features most influenced the model's predictions. As illustrated in Figure 2, capital gain emerged as the dominant predictive feature. Notably, gender appeared consistently among the top ten influential

features, suggesting the model has learned a dependence on this protected attribute despite it playing no legitimate role in income prediction.

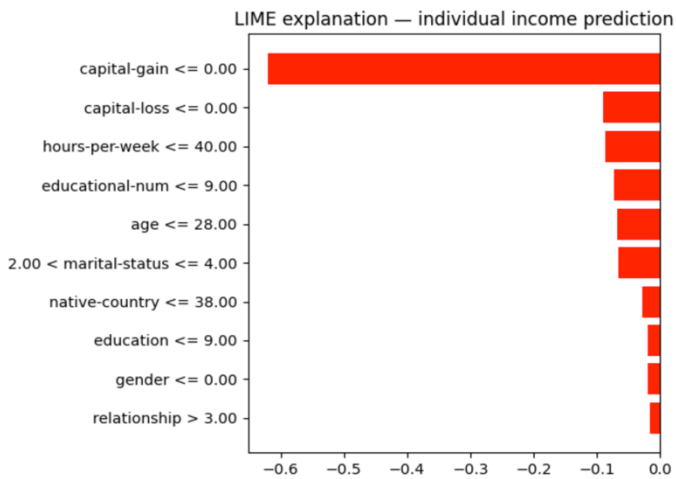

Figure 2 LIME feature importance scores for a single instance. All features push the prediction towards ≤50K. Gender appears as a contributing factor.

DiCE counterfactuals were generated for the same instance to identify what minimal changes would flip the prediction to >50K. As shown in Figure 3, all three counterfactual scenarios required substantial increases in capital gain, ranging from approximately 89,000 to 99,000. One scenario additionally suggested a change in occupation and another in education level. Importantly, gender, age and race were not modified in any of the three counterfactuals, meaning DiCE avoided non-actionable recourse for this instance.

1 row · 1 rows × 15 cols · Static Output

	age	workclass	fnlwgt	education	educational-num	marital-status	occupation	relationship	race	gender	capital-gain	capital-loss	hours-per-week	native-country	income
0	19	4	30800	0	6	3	0	4	0	0	0	0	40	38	0

Diverse Counterfactual set (new outcome: 1)

3 rows · 3 rows × 15 cols · Static Output

	age	workclass	fnlwgt	education	educational-num	marital-status	occupation	relationship	race	gender	capital-gain	capital-loss	hours-per-week	native-country	income
0	-	-	-	-	-	-	12	-	-	-	92035	-	-	-	1
1	-	-	-	-	-	-	-	-	-	1	99265	-	-	-	1
2	-	-	-	2	-	-	-	-	-	-	89848	-	-	-	1

Figure 3 DiCE counterfactual suggestions for a 19-year-old instance predicted as ≤50K. To flip the prediction to >50K, all three scenarios require substantial increases in capital gain (89k–99k), with one additionally suggesting a change in occupation (to 12) and another a change in education (to 2). Gender, age, race and most other features remain unchanged across all counterfactuals.

To directly address the research question, a counterfactual flip rate analysis was conducted by modifying only the gender attribute across the entire test set. As shown in Figure 4, this single change caused the model to alter its prediction for 480 instances, yielding a flip rate of 5.31%.

This serves as a diagnostic of implicit gender bias and the model treats gender as a meaningful signal despite it being a protected attribute.

```python
# How often does changing ONLY gender flip the prediction?
# This is our bias diagnostic — not a recommendation

X_test_gender_flipped = X_test.copy()

# Flip gender: 0 becomes 1 and 1 becomes 0
X_test_gender_flipped['gender'] = 1 - X_test_gender_flipped['gender']

# Get predictions for original and gender-flipped versions
original_preds = rf_model.predict(X_test)
flipped_preds = rf_model.predict(X_test_gender_flipped)

# Count how many predictions changed purely due to gender flip
n_flipped = (original_preds != flipped_preds).sum()
flip_rate = n_flipped / len(original_preds)

print(f"Total test instances: {len(original_preds)}")
print(f"Predictions changed by gender flip: {n_flipped}")
print(f"Counterfactual flip rate (gender only): {flip_rate:.2%}")
```

✓ [10] 417ms

```
Total test instances: 9045
Predictions changed by gender flip: 480
Counterfactual flip rate (gender only): 5.31%
```

Figure 4 Gender-only counterfactual flip rate across 9,045 test instances. 5.31% of predictions changed when solely gender was modified.

Finally, an actionability analysis was conducted across 10 test instances and 30 counterfactuals. As shown in Figure 5, 83% of DiCE's suggestions involved actionable features such as capital gain and education, while 17% included non-actionable attributes such as age and race, confirming that standard DiCE does not natively constrain for fairness (Goethals et al., 2023).

All code is available at: https://github.com/JoJmH/XAI-report (Jomah, 2026)

Discussion

Several limitations were observed across both methods. Initially, LIME's instability means feature attributions may shift across runs due to its reliance on random sampling, meaning the same individual could receive a different explanation each time (Garreau & von Luxburg, 2020). Next, while DiCE was mostly successful at leaving protected attributes alone, 17% of its suggested changes still involved fixed traits like age and race, confirming that the standard version of DiCE does not enforce fairness constraints (Goethals et al., 2023). Furthermore, DiCE

occasionally produced completely impractical advice. For instance, it suggested that a 19-year-old increase their capital gains by nearly 100,000. This highlights a core problem with counterfactuals because optimising for the shortest distance to a new decision does not ensure the resulting suggestions make sense in the real world. Lastly, both techniques focus purely on explaining the model rather than questioning if an algorithm should be entrusted with such decisions at all. This poses a significant ethical concern in critical fields like finance and human resources, where affected individuals frequently lack the power to push back against hidden automated processes (Wachter et al., 2017).

Ultimately, despite these limitations, these findings suggest that LIME and DiCE act as complementary tools for diagnosing model bias. While LIME exposes how heavily the model weighs gender for individual predictions, DiCE uses a 5.31% counterfactual flip rate to measure exactly how often that bias alters the final outcome. Neither approach is sufficient in isolation. LIME cannot illustrate how easily a decision can be reversed, and DiCE cannot directly isolate the problematic feature weights driving the model's logic.
Taken together, this study demonstrates that counterfactual flip rate analysis via DiCE provides a more direct and quantifiable diagnostic of gender bias than LIME's feature attributions alone, directly addressing the research question posed.

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